NOV 25 1983

MIL-F-16377/59A(SH)
9 May 1983
SUPERSEDING
MIL-F-16377/59(SHIPS)
23 August 1972

MILITARY SPECIFICATION SHEET

FIXTURES, LIGHTING; FLUORESCENT, DETAIL LIGHTING FOR STEP ILLUMINATION

(A) 8 WATTS, 120 VOLTS, 60 HERTZ, WATERTIGHT, SYMBOL 353, 353.1 AND 353.2

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

(A) The complete requirements for acquiring the fixtures described herein shall consist of this document and the latest issue of MIL-F-16377(SH).

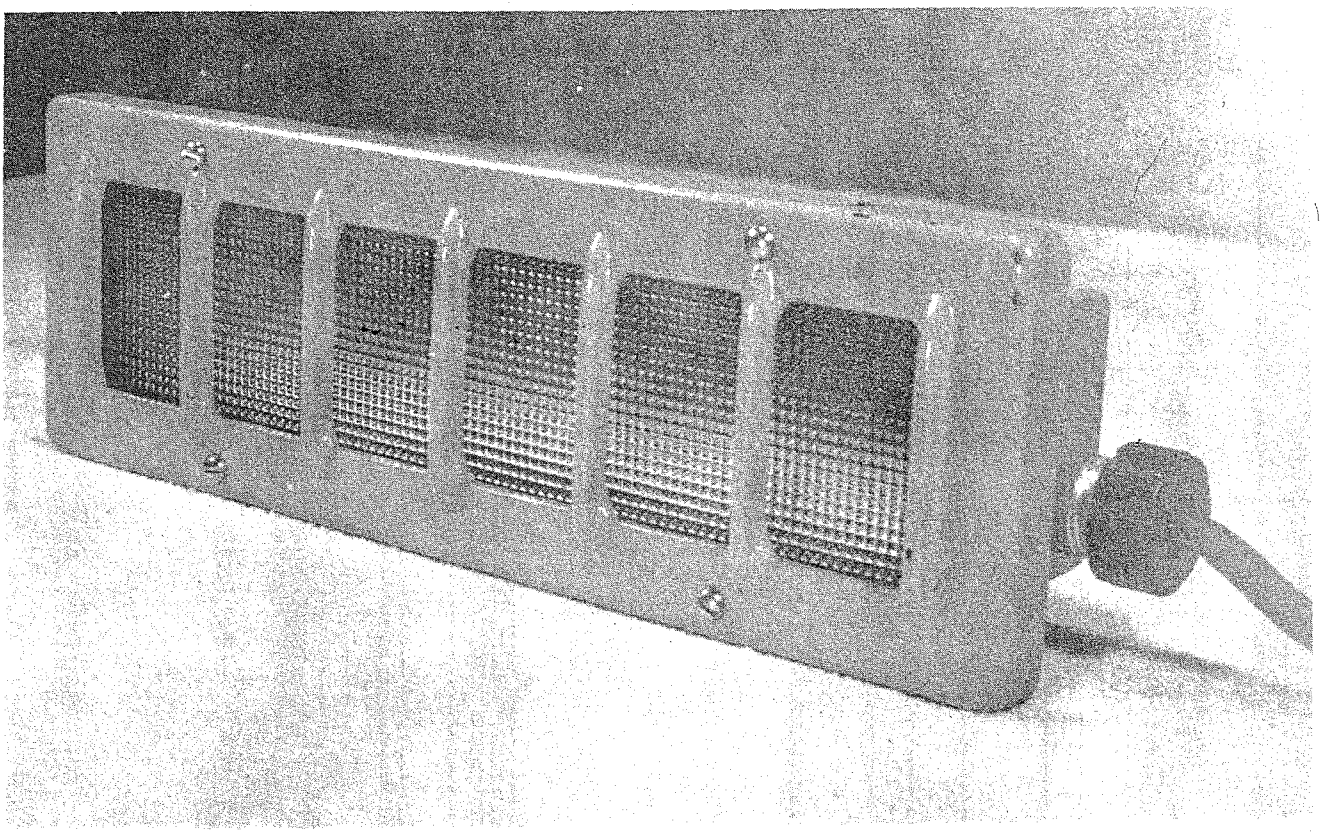


FIGURE 1. Typical view of Symbols 353 and 353.1.

All dimensions in inches.
For tolerances, other than specified herein, see paragraph 3.5.2 of MIL-F-16377.

(A) Denotes changes

FSC 6210

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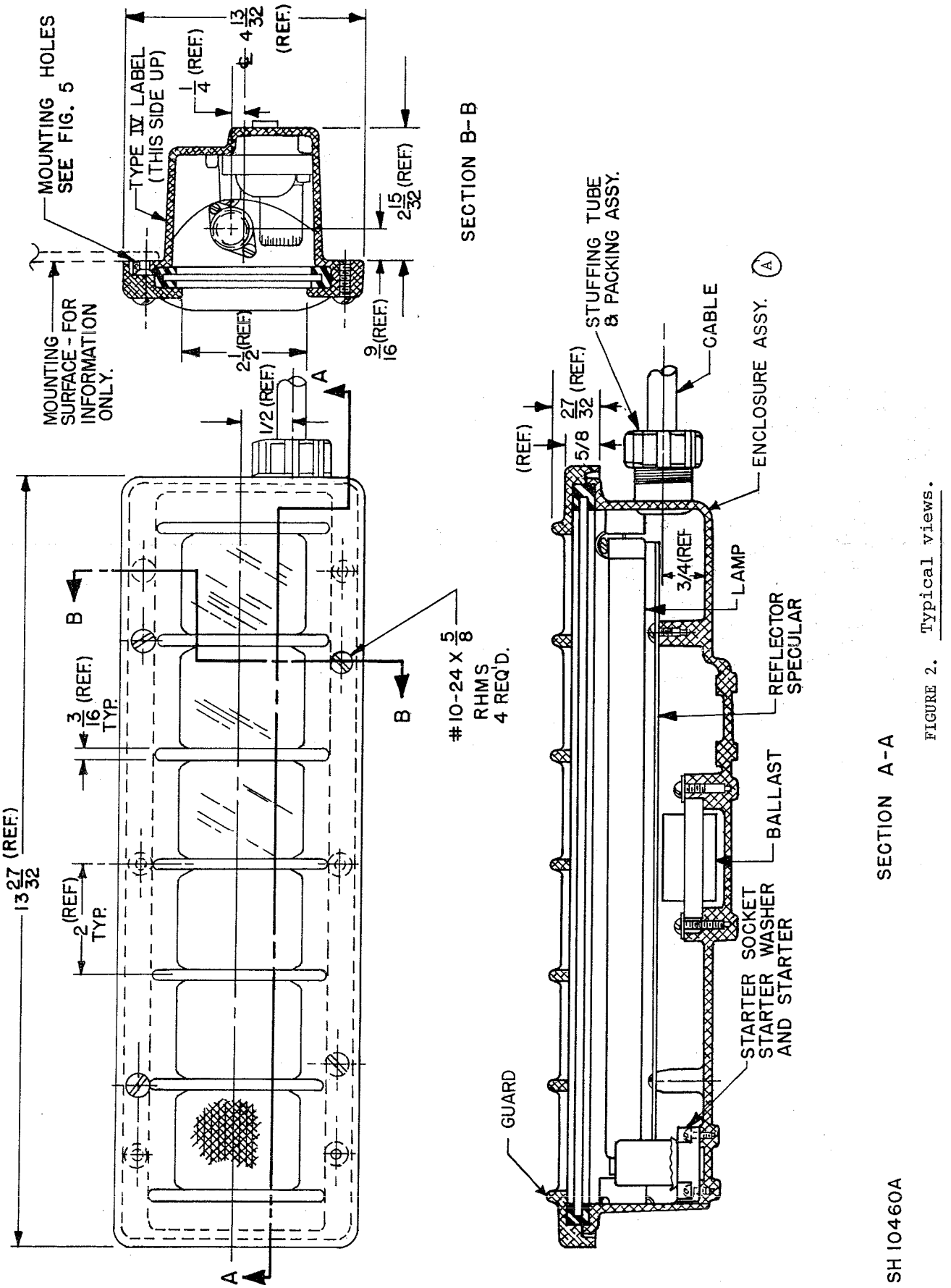
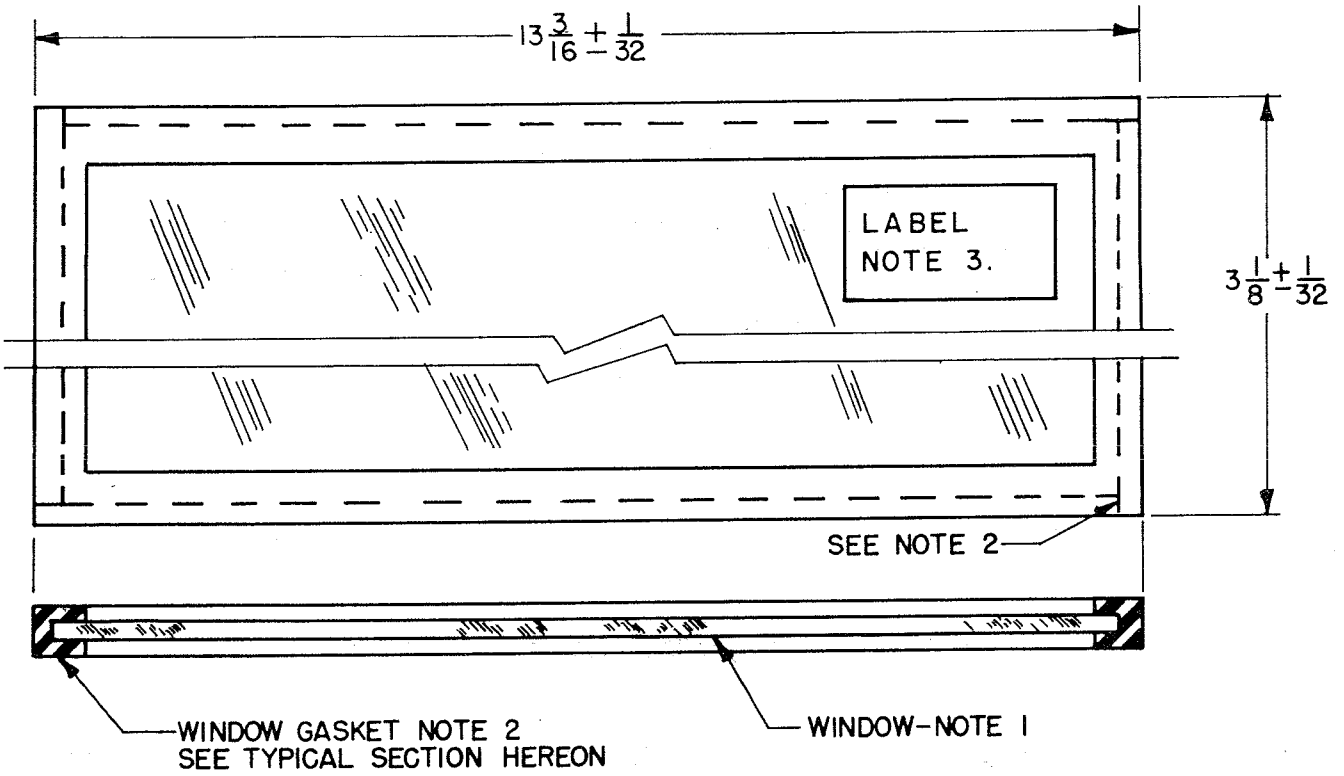


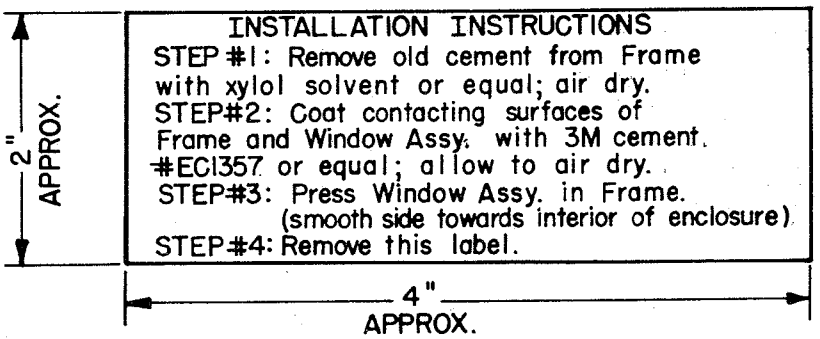
FIGURE 2. Typical views.

SH 10460A

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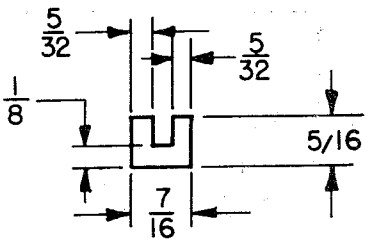


(a) Window Assembly



SH 10461

(b) LABEL - Note 3.



STANDARD RMA TOLERANCES FOR EXTRUSION SHALL APPLY

(c) GASKET-Typical Section

NOTES:
1. Material for window shall be acrylic plastic as follows:
   Part Number M16377/59-001 for Symbol 353: Clear, prismatic (one side only), 1/8 inch thick, type CLEAR-2 of MIL-P-24191.
   Part Number M16377/59-002 for Symbol 353.1: Red, prismatic (one side only), 1/8 inch thick, type RTPS-2 of MIL-P-24191.
   Part Number M16377/59-003 for Symbol 353.2: Blue, prismatic (one side only), 1/8 inch thick, type BTPS-2 of MIL-P-24191.
2. Material for window gasket shall be neoprene rubber 25-35 durometer. Gasket shall be cemented to window. Gasket may be furnished in segments.
3. For window assemblies furnished as repair parts, a type IV label shall be furnished in accordance with MIL-F-16377. Location and the information to be shown on this label shall be as shown hereon.
4. Window assembly shall be cemented to window frame.

FIGURE 3. Window assembly.

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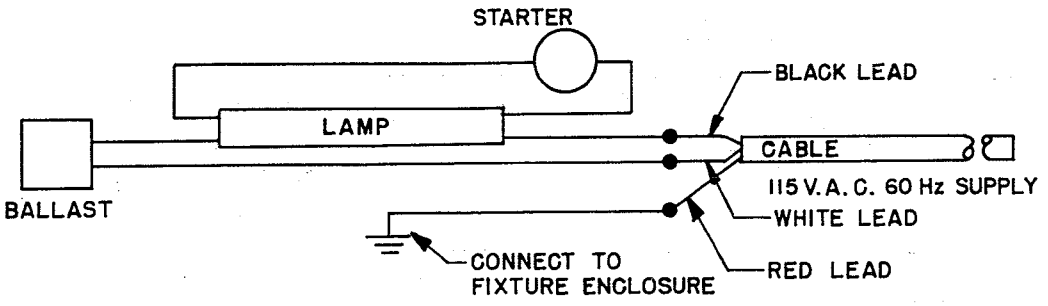
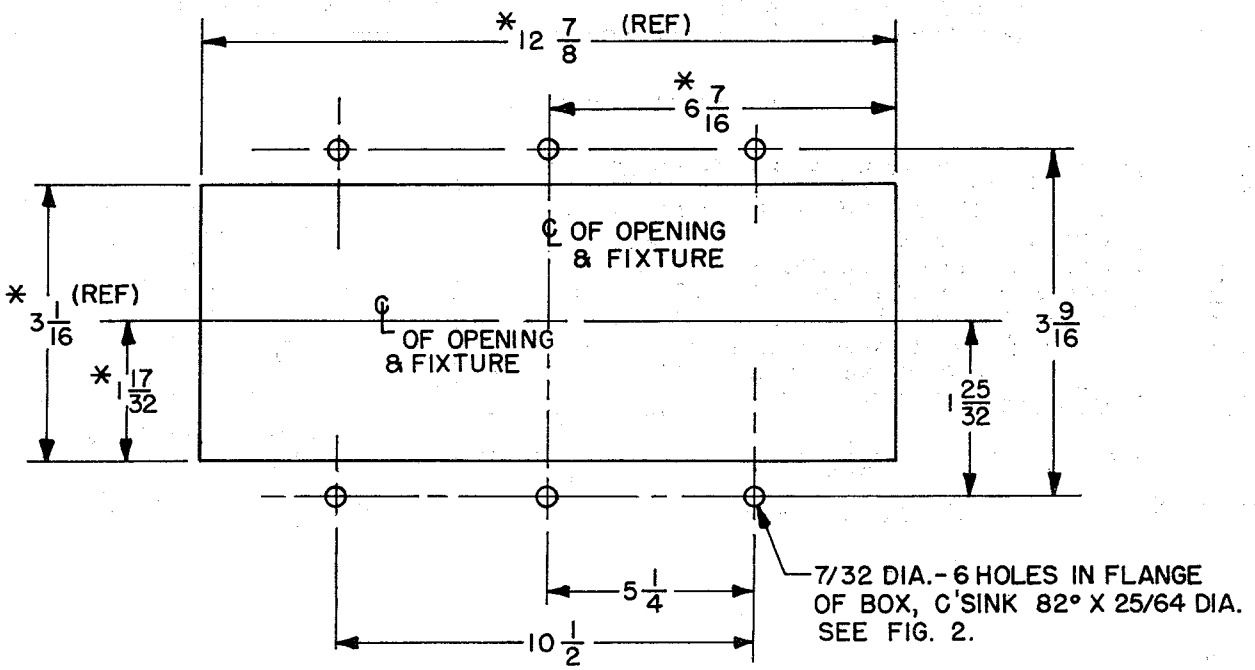


FIGURE 4. Wiring Diagram.

* DIMENSIONS SHOWN ARE FOR OPENING IN MOUNTING SURFACE

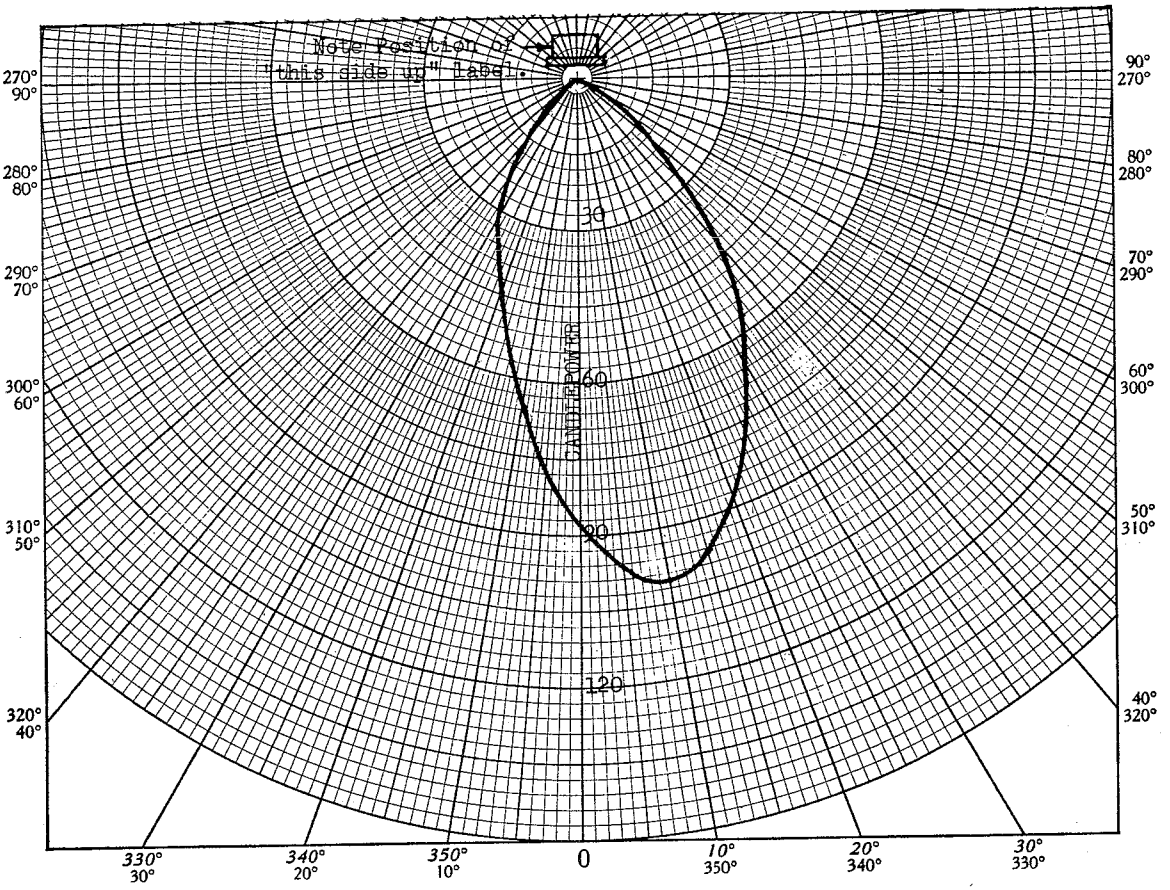


MOUNTING BOLTS SHALL BE # 10-24 FLAT HEAD MACHINE SCREWS (TO BE FURNISHED BY INSTALLING ACTIVITY.)

SH 10462

FIGURE 5. Mounting dimensions.

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Photometric test data:

Distance - - - 10 feet.
Plane - - - - At right angles to long axis of fixture through geometric center of lamp.
Voltage - - - 115 volts, 60 Hz.
Ambient temperature - - - 76°F.
Lamp: Use lamp furnished with fixtures seasoned for at least 100 hours.
Ballast: Use ballast furnished with fixture.
Lamp Correction: - - - Distribution curves based on 380 lumens per lamp.

FIGURE 6. Candlepower distribution curves.

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REQUIREMENTS:
Type: I
Class: 1
Symbol 353: Complete lighting fixture with one 8-watt cool white lamp and clear prismatic lens assembly.
Design:
(A) Electrical rating: 120 volts, 60 hertz, 8 watts.
Dimensions and configuration: See figures 1 and 2.
Cable: Ten feet of type SJO or SJT cable, AWG size 16, three conductor (black, white, and red), in accordance with J-C-580.
Window assembly: See figure 3.
Wiring diagram: See figure 4.
Mounting dimensions: See figure 5.
Light output: See figure 6 (candlepower).
Enclosure: Watertight. The enclosure shall be made of 3/32 inch minimum thickness cast aluminum.
Weight: 5 pounds maximum.
Labels: Labels shall be furnished in accordance with the following types of MIL-F-16377:
(a) Type I
(b) Type IV - One type IV label shall be applied as shown on figure 2. The information to be shown on the type IV label shall be "THIS SIDE UP".
Standard parts: The standard parts shown in table I shall be used in the construction of the fixtures specified herein.

TABLE I. Standard parts.

Name	Specification of Military part number	National Stock Number (NSN)	Remarks
Lamp	MS15588-3	6240-00-299-5546	Industry number F8T5/CW
Lamplock	M16377/40-001	6250-00-399-7039	-------
Lampholder	MIL-L-970/10	6250-00-377-8590	-------
Startersocket	MIL-L-970/13	6250-00-295-2738	-------
(A) Starter	8W, type III of W-S-755	6250-00-299-5962	Commercial Designation: FS5
Ballast	M16377/45-001	6250-00-295-2735	-------
Red filter tube	M16377/42-001	6210-00-399-6297	Not furnished See paragraph 3.7.7 of MIL-F-16377.
Blue filter tube	M16377/42-002	----------	Not furnished See paragraph 3.7.7 of MIL-F-16377.

Quality assurance: Quality assurance shall be as specified in MIL-F-16377 and as follows:
First article inspection:
Complete fixtures: See table II.
Window assemblies: See table III.
Quality conformance:
Comparison inspection: Same as the first article inspection.
Inspection of product for delivery:
Examination: See MIL-F-16377.
Examination for preparation of delivery: See MIL-F-16377.
Quality conformance tests: See MIL-F-16377.

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TABLE II. First article and comparison inspection for complete fixtures.

Tests	Requirement paragraph of MIL-F-16377	Test paragraph of MIL-F-16377	Remarks
Examination	3.13	4.5	-------
Operation	3.13	4.8.1	-------
Dielectric withstanding voltage	3.6.15	4.8.2	-------
Insulation resistance	3.6.16	4.8.3	-------
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Shock	3.5.12	4.8.7	Grade B
Vibration	3.5.13	4.8.8	-------
Enclosure effectiveness	3.5.11	4.8.14.4	Watertight
Dielectric withstanding voltage	3.6.15	4.8.2	-------
Insulation resistance	3.6.16	4.8.3	-------
Noise	3.5.14	4.8.9	-------
Grounding circuit	3.6.13.3	4.8.11	-------
Optical uniformity	3.7.4.1.1	4.8.4	-------
Light output	3.7.1	4.8.6.1	Candlepower
Magnetic permeability	3.4.8	4.8.18	-------

TABLE III. First article and comparison inspection for window assemblies.

Tests	Requirement paragraph of MIL-F-16377	Test paragraph of MIL-F-16377	Remarks
Examination	3.13	4.5	-------
Optical uniformity	3.7.4.1.1	4.8.4	-------

General information: (a) Assignment of Military part numbers and general information is shown in table IV.

TABLE IV. Part number and general information.

Military part number M16377/59-	Item name	National Stock Number (NSN)	Remarks
353	Complete fixture, Symbol 353	Not yet assigned	-------
001	Window assembly	Not yet assigned	See figure 3

(b) This light is intended for standing illumination in berthing areas (red, CIC spaces (Blue), and at the foot of ladders.

Preparing activity:
Navy - SH
(Project 6210-N 590)

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